

EX.NO : 02

ANALYZING THE PERFORMANCE OF VARIOUS CONFIGURATIONS AND PROTOCOLS OF LAN ESTABLISHING A LOCAL AREA NETWORK (LAN)

AIM :

To set up a Local Area Network using Cisco Packet Tracer.

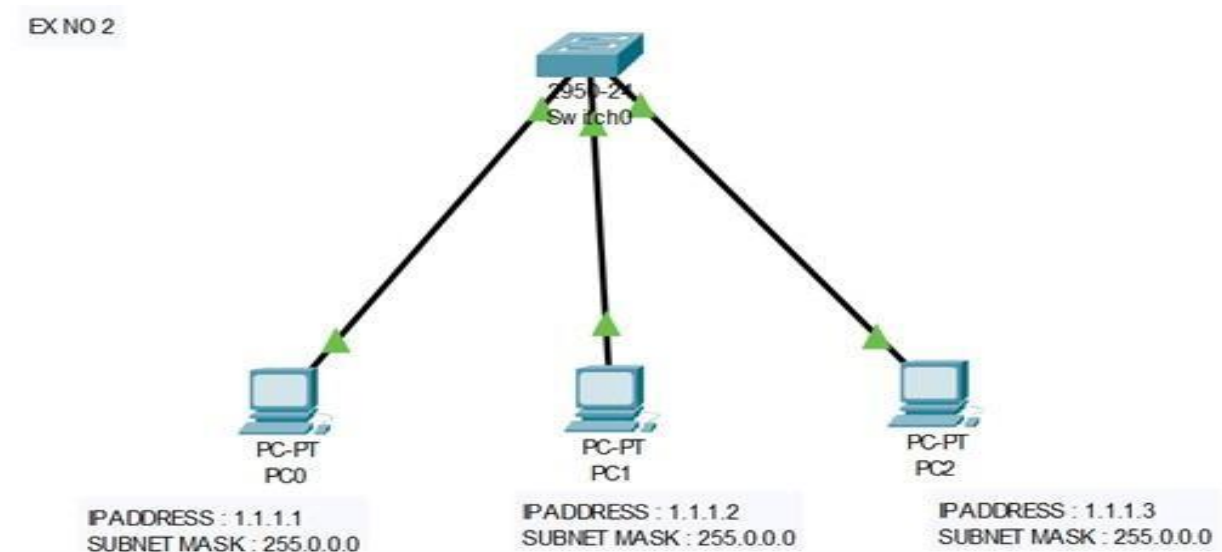
REQUIREMENTS :

- Three Windows PC.
- One Switch(2950-24) or One Hub.
- Three Straight Line LAN Cables.
- Cisco Packet Tracer.

PROCEDURES :

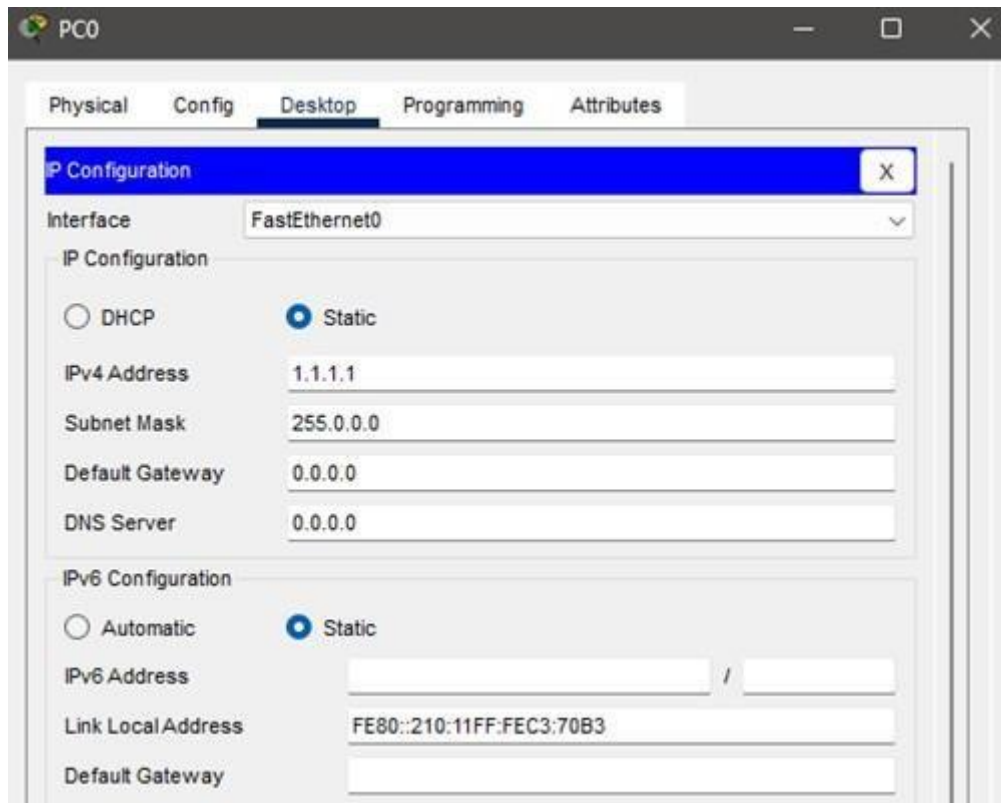
- Open CISCO PACKET TRACER software.
- Draw the Three PC using END Device Icons.
- Draw the Cisco 24 Port Switch Using Switch icon lists.
- Make the Connections using Copper-Straight-Through Ethernet Cables.
- Enter the IP Address To Each Machine.
- Check the Network Connections using Add Simple PDU(P).

NETWORK TOPOLOGY :



HOST IP ADDRESS :

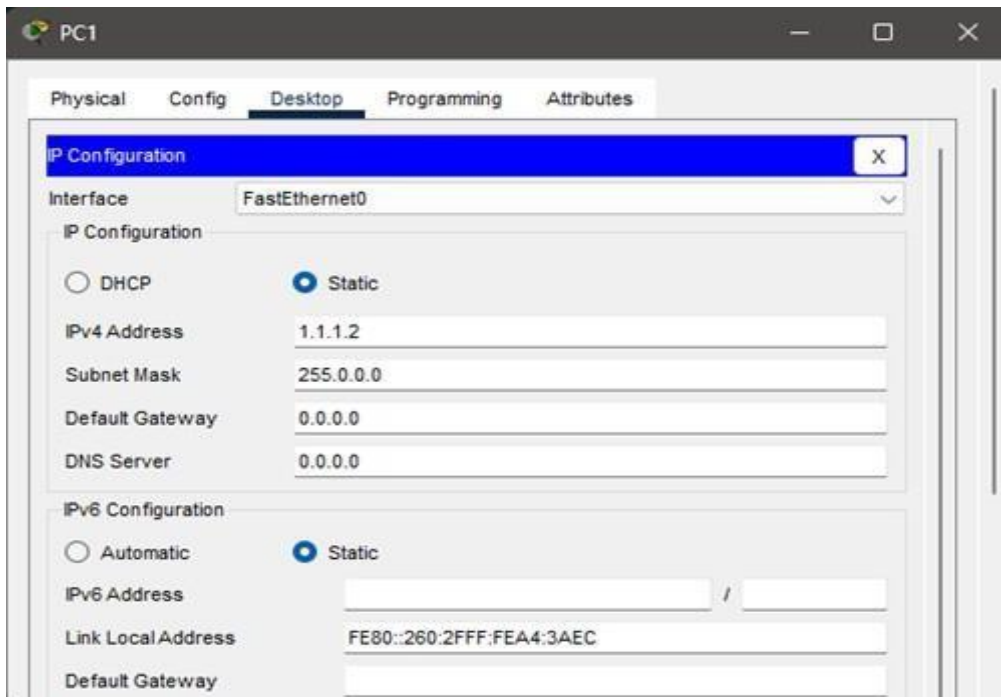
PC0 IP CONFIGURATION



The image shows the 'PC0' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The 'IPv4 Address' is '1.1.1.1', 'Subnet Mask' is '255.0.0.0', 'Default Gateway' is '0.0.0.0', and 'DNS Server' is '0.0.0.0'. Under 'IPv6 Configuration', the 'Static' radio button is selected. The 'IPv6 Address' field is empty, 'Link Local Address' is 'FE80::210:11FF:FEC3:70B3', and 'Default Gateway' is empty.

Field	Value
Interface	FastEthernet0
IP Configuration	Static
IPv4 Address	1.1.1.1
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	Static
IPv6 Address	
Link Local Address	FE80::210:11FF:FEC3:70B3
Default Gateway	

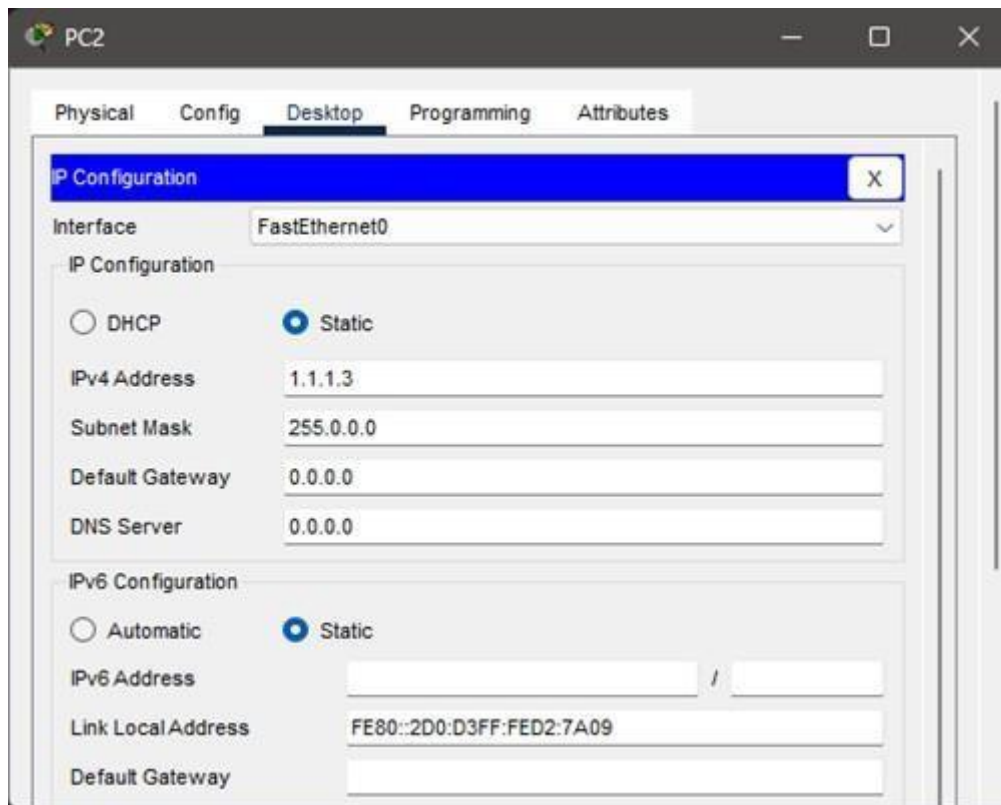
PC1 IP CONFIGURATION



The image shows the 'PC1' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The 'IPv4 Address' is '1.1.1.2', 'Subnet Mask' is '255.0.0.0', 'Default Gateway' is '0.0.0.0', and 'DNS Server' is '0.0.0.0'. Under 'IPv6 Configuration', the 'Static' radio button is selected. The 'IPv6 Address' field is empty, 'Link Local Address' is 'FE80::260:2FFF:FEA4:3AEC', and 'Default Gateway' is empty.

Field	Value
Interface	FastEthernet0
IP Configuration	Static
IPv4 Address	1.1.1.2
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	Static
IPv6 Address	
Link Local Address	FE80::260:2FFF:FEA4:3AEC
Default Gateway	

PC2 IP CONFIGURATION

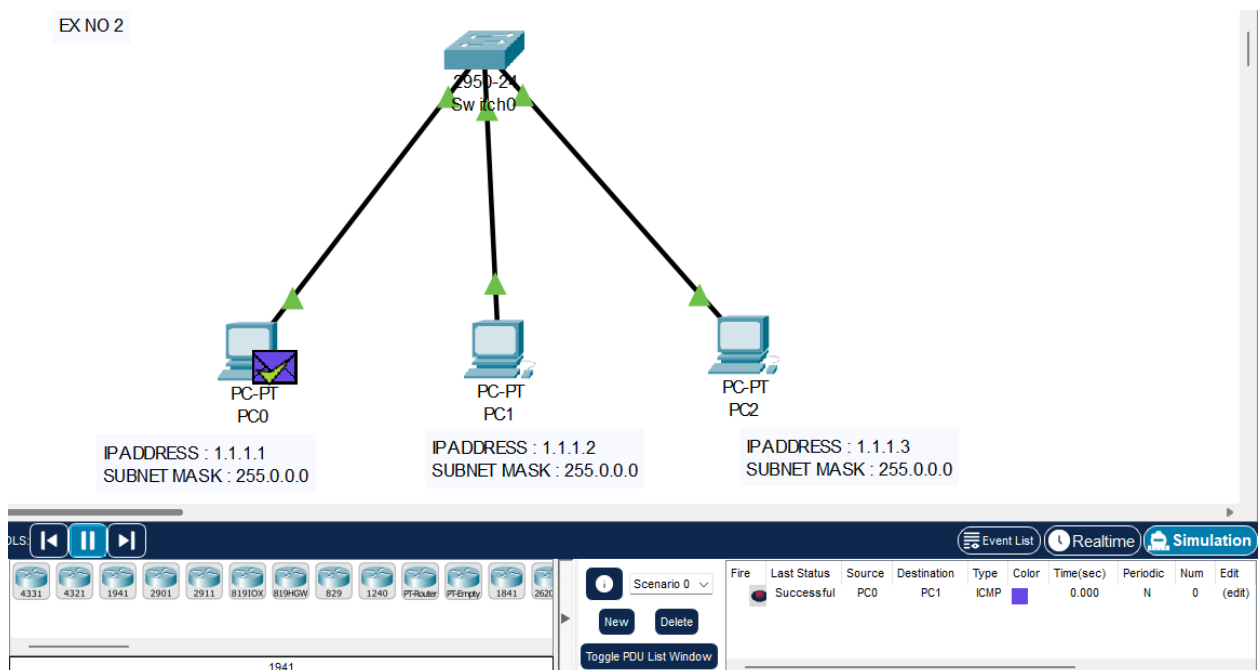


VERIFY LAN NETWORK CONNECTIVITY :

Using Add Simple PDU(p), Click the mail icon and then drop one mail to PC0 and another mail to PC1. If the resultant window shows the successful delivery, then network connectivity is successful.

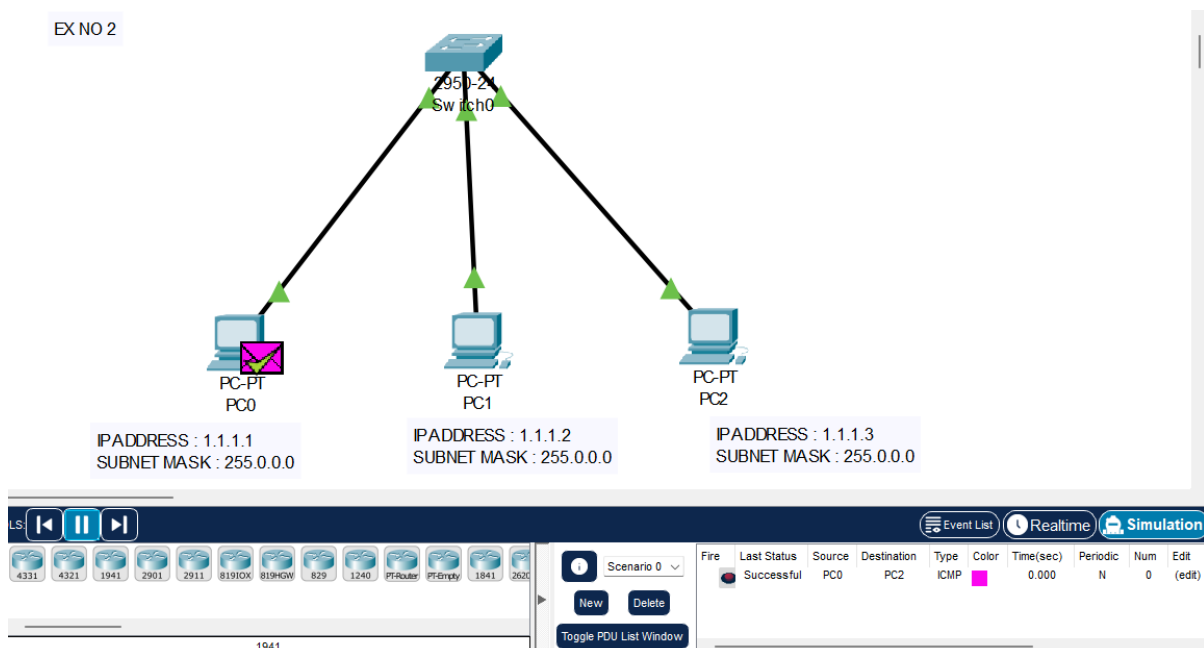
HOST PC0 TO PC1

EX NO 2



HOST PC1 TO PC2

EX NO 2



RESULT :

Thus, the LAN connection is established, hosts are configured, the communications among the machines are verified and manipulated successfully.